

Vishnusmaran B Rai

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Objective

Computer Science Engineering student with practical experience in full-stack development, computer vision, and applied cryptography. Proven ability to build secure, data-driven applications, backed by strong organizational and financial management skills demonstrated as a CSE Department leader. Seeking an engineering role to contribute to scalable software solutions.

Education

B.E. in Computer Science: Alva's Institute of Engineering and Technology - CGPA: 6.88 (Pursuing)

PUC: Boscross PU College, Mangalore, Karnataka - 78.83% 2021–2023

SSLC: Lourdes Central School, Mangalore, Karnataka - 77.66% 2021

Projects

Smart Hotel Monitoring System

Tech Stack: Python, FastAPI, YOLO, OpenCV

Repository: github.com/vinith-0430/Smart-hotel-monitoring-system

Developed an automated vision-based surveillance system to maintain high culinary standards. The system leverages YOLO for real-time detection of hygiene protocol breaches and pest presence, instantly notifying management via SMTP-integrated email alerts containing visual evidence for immediate intervention.

SecureVault

Tech Stack: Python, Streamlit, SQLite, Cryptography

Repository: github.com/virajj12/securevault

Engineered a privacy-focused data storage platform utilizing AES-256 encryption and a robust audit logging system. Implemented advanced cryptographic techniques, including HMAC-based blind indexing, to allow secure database querying while ensuring sensitive information remains encrypted at all times.

Notes RAG

Tech Stack: Python, LLMs, OCR

Repository: github.com/rai-vishnu965/notes-rag

Developed a Retrieval-Augmented Generation (RAG) application to intelligently query and retrieve information from personal notes. Built custom pipelines for document ingestion featuring Optical Character Recognition (OCR), alongside modules for text embedding, vector indexing, and user querying.

Technical Skills & Certifications

- **Languages:** Java, C, SQL, Python
- **Web Development:** HTML5, CSS3, JavaScript, FastAPI, Streamlit
- **Tools & Frameworks:** OpenCV, YOLO, SQLite, Git
- **Certifications:** Infosys Springboard - Associate in IT Foundation Skills (Java), DSA, DBMS, NoSQL
- **Upskilling:** Salesforce Trailhead (In Progress)

Extracurricular Activities

- **Treasurer, Xypheria - National Level Hackathon:** Managed the budget, sponsorship, expenses, and allocated funds efficiently across different event tracks.
- **Treasurer, CSE Department (2025–26):** Handled the department's financial portfolio, accurately managing budgets, auditing event expenses, and maintaining comprehensive monetary records.
- **Organizing Committee Member, TEDx AIET:** Collaborated within a cross-functional team to plan, coordinate, and execute operational logistics for the event, ensuring a high-quality attendee experience.

- **Hackathon Competitor:** Actively participated in competitive hackathons, collaborating in fast-paced team environments to brainstorm, prototype, and present software solutions under tight timelines.